

Question

Halogen chemistry is an important part of elemental chemistry, and **X** element is a halogen. The simple ion X^+ is often considered unable to exist independently, but it was later discovered that compound **A** containing a monovalent positive **X** element can be obtained under the following conditions: $\text{X}_2 + \text{B} + \text{C} \rightarrow \text{A}$. Known: **B** is a common silver source, and its anion does not contain halogens and is an ionic cluster; **C** is a common base in organic synthesis and has a certain nucleophilicity, and is considered to be used as a nucleophilic catalyst. The molecular structure of **C** has multi-center delocalized bonds and contains only one *N* atom; oxidation-reduction and acid-base complexation reactions occur during the reaction to generate **A**.

Based on the above information, determine the correct option from the following statements.

- A.** **X** is the element bromine, **C** reacts with elemental bromine, Br_2 acts as an electrophile and is attacked by the nitrogen atom of **C**.
- B.** **A**'s monovalent **X** combines with one molecule of **C**, thereby obtaining a stable anion which is nitrate.
- C.** **X** is an iodine atom, because the electronegativity of the iodine atom is relatively small, and it combines with **C** in a 1:1 ratio.
- D.** **B** is silver nitrate, which is often used to test for halide anions. In the synthesis of **A**, the nitrate acts as an oxidizing agent.
- E.** The ratio of **X** atoms to **C** molecules in **A** is 1 to 2, because each **C** molecule contains one *N* atom, so the ratio of **X** atoms to *N* atoms is also 1 to 2. In addition to **A**, the reaction also produces a product containing monovalent anionic **X**.
- F.** Hypervalent bonding exists in **A**.

Answer

Correct Answer: **F**

Detailed Explanation

Halogen elements have high electronegativity and are difficult to form X^+ . Therefore, **X** should be the *I* element with the largest radius and the smallest electronegativity;

CHECKPOINT

1 PTS

X : *I*

B is a common silver source, and the anion does not contain halogens, so it should be silver nitrate (AgNO_3). Silver nitrite is uncommon;

CHECKPOINT

1 PTS

B: AgNO_3

C is a common base in organic chemistry that can be used as a nucleophilic catalyst and has only one nitrogen atom. It can be determined as pyridine ($\text{C}_5\text{H}_5\text{N}$);

CHECKPOINT

1 PTS

C: C_5H_5N

I^+ has one empty orbital and three lone pairs of electrons. If it is 1:1 coordination, it will form an N-I single bond, which is unstable and has too high energy. If it is 1:2 coordination, the empty orbital of I forms a 3-center 4-electron bond with the lone pair electrons of N on two pyridines, which is more stable than a single bond. So **A** is $[I(C_5H_5N)_2]NO_3$

CHECKPOINT

1 PTS

A: $[I(C_5H_5N)_2]NO_3$

According to the above analysis, the correct statement in each option should be F.

The element in A is wrong, and the reaction mechanism is complex, not a simple nucleophilic attack reaction, which is wrong.

The ratio of X to pyridine in B is wrong.

The ratio of X to pyridine in C is wrong.

In D, nitrate does not act as an oxidant, which is wrong.

In E, nitrate also has N atoms, so the ratio of **X** to N is 1:3.